Beyond the Opening Weekend: Measuring Sustained Audience Engagement in Disney Content

A Persistence Score Analysis of Franchise vs. Original Films (n=24)

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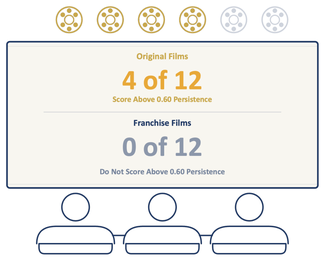
Background / Problem to Solve

Disney produces two types of content: original films (standalone stories) and franchise films (sequels and spin-offs based on existing IP). Conventional wisdom assumes franchise films are safer investments due to built-in audience recognition and stronger opening weekends. But does initial popularity translate to lasting impact? This project challenges that assumption by asking: what drives sustained audience engagement peak popularity, or persistence?

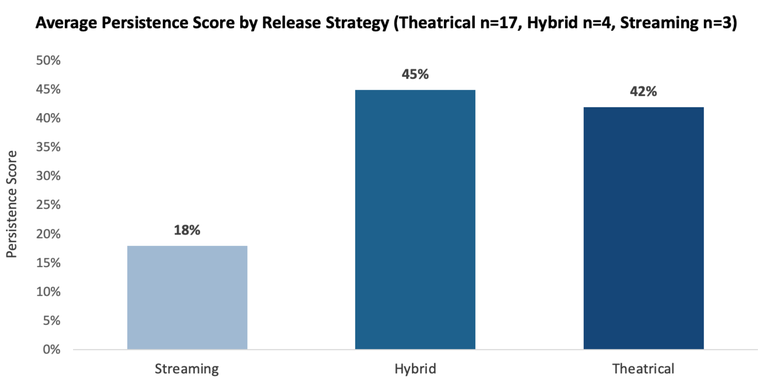


Data & Methods

Dataset: 24 Disney/Pixar/Marvel films (2014–2023) · n=12 franchise, n=12 original Data Sources: Google Trends (weekly search interest, U.S., 12-week window per film) · TMDB API (popularity, vote average, vote count) Custom Metric: Persistence Score a behavioral measure of sustained audience attention derived from Google Trends (see formula box) Statistical Tests: Mann-Whitney U (primary, non-parametric, one-tailed) + Welch's t-test (secondary, confirmatory) Additional Analyses: IQR outlier detection · Shapiro-Wilk normality test · Sensitivity check (Soul excluded)

Persistence Score Formula

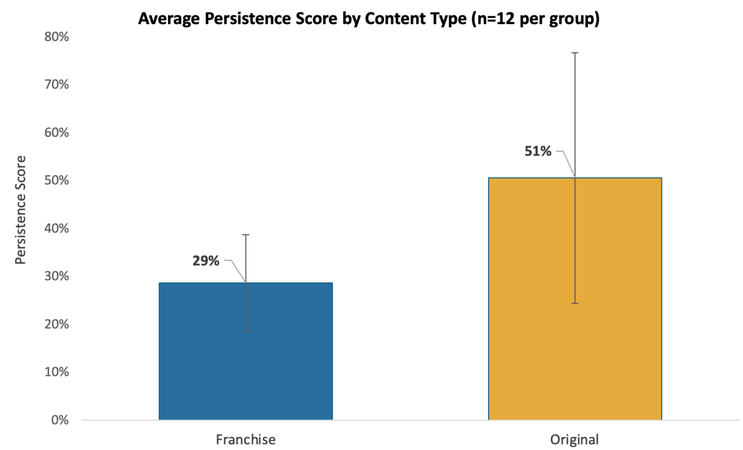
Persistence Score = Avg Google Trends Interest (Weeks 5–12) ÷ Avg Google Trends Interest (Weeks 1–4) Peak window = weeks 1–4 · Sustained window = weeks 5–12 Measures sustained audience attention after initial release. Higher score = stronger long-term retention. Range: 0 → 1. Component weights: Half-Life (post-peak) 33.3% · Post-Peak Decay Rate (8w) 33.3% · Total Engagement (12w) 33.3% *Floor adjustment (×0.5) applied exclusively to Soul: COVID-era contextual anomaly, sensitivity-tested separately.*



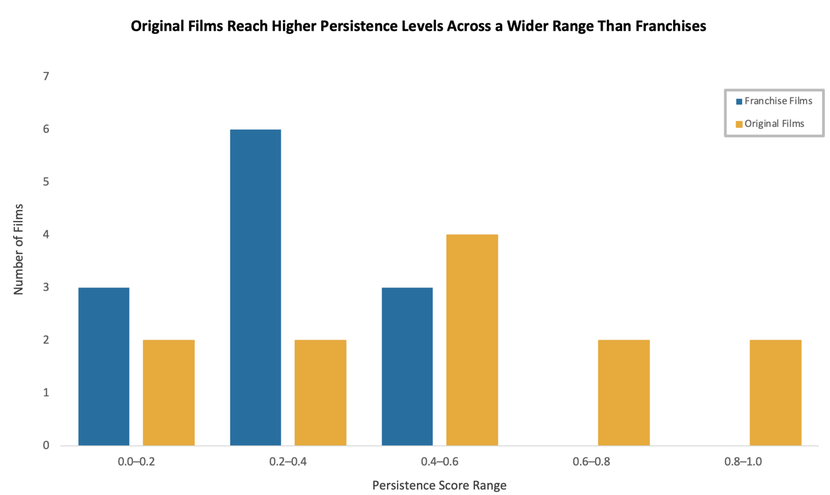
- Streaming's lower persistence (0.18) reflects COVID-era context, not content quality, all three streaming releases (Luca, Turning Red, Soul) were Disney+ exclusives during 2020–2022.

Results

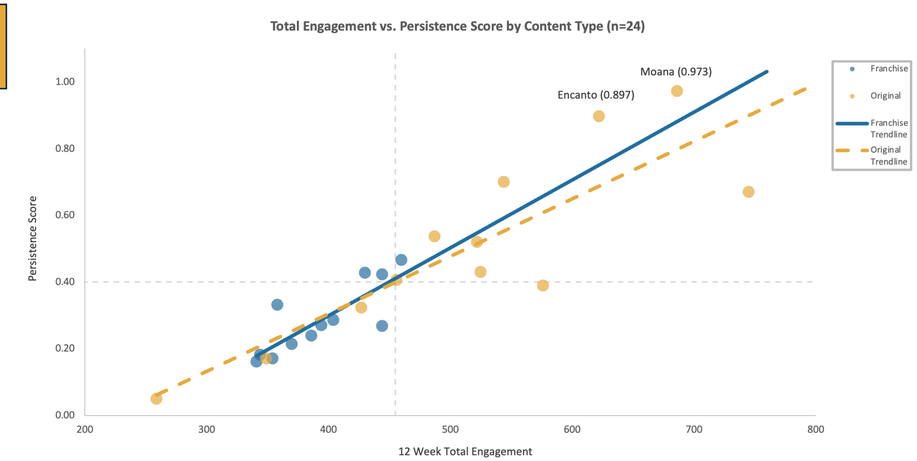
Original films score 76.3% higher on the Persistence Score than franchise films (mean: 0.506 vs. 0.287, n=12 per group). Mann-Whitney U test: U=33, p=0.013 (one-tailed), r=0.54 (large effect). Confirmatory Welch's t-test: t=2.59, p=0.011, Cohen's d=1.11 (large effect). Both tests independently significant, convergent evidence across parametric and non-parametric methods. Sensitivity check excluding Soul (COVID-era outlier): p=0.003, r=0.68 (highly significant).



- Error bars = ±1 std dev. Franchise films cluster consistently (std=0.10); originals show high variance (std=0.26): a high-risk, high-reward engagement pattern.



- Original films show higher score variability (std=0.26 vs. franchise std=0.10), reflecting a high-risk, high-reward pattern, some originals sustain exceptionally (Moana, Encanto) while others decay quickly.
- Franchise films have zero films scoring above 0.60. Originals account for all four films in the upper range, including Moana (0.973) and Encanto (0.897).



- Both groups show a positive relationship between total engagement and persistence. Originals deliver stronger long-term retention at higher engagement levels (Original R^2=0.74, p<0.001; Franchise R^2=0.67, p=0.001).

Analysis & Conclusions

This analysis challenges the conventional franchise-first model in studio content strategy. While franchise films generate faster initial attention, original content demonstrates consistently stronger sustained engagement, suggesting that long-term retention and opening weekend performance are fundamentally different measures of success.
Original films show higher score variability (high-risk, high-reward) while franchise films are more consistently moderate, with direct implications for content investment: originals carry more variance but higher upside. For Disney, sustained engagement drives streaming retention, merchandise longevity, and cultural staying power, revenue streams that extend well beyond opening weekend.
The Persistence Score captures a dimension of audience behavior distinct from platform popularity, vote counts, and critical ratings, confirming that search-based sustained interest measures something traditional metrics miss.

Limitations

Sample limited to 24 Disney/Pixar/Marvel films, results directional, not causal. Google Trends provides relative, not absolute, search interest. No direct streaming data available; engagement approximated through publicly observable search behavior. TMDB outcome variables reflect theatrical scale and critical reception rather than long-term engagement, limiting their use as independent validators.

References

Mann, H.B. & Whitney, D.R. (1947). Annals of Mathematical Statistics, 18(1), 50–60. Google LLC. (2024). Google Trends [Search interest data]. trends.google.com The Movie Database (TMDB). (2024). TMDB API [Film metadata]. themoviedb.org Kerby, D.S. (2014). Comprehensive Psychology, 3, 11.